Prototype and Context-Enhanced Learning for Unsupervised Domain Adaptation Semantic Segmentation of Remote Sensing Images

Kuiliang Gao, Anzhu Yu, Xiong You, Chunping Qiu, and Bing Liu

***Abstract*— In unsupervised domain adaptation (UDA) of remote sensing images (RSIs), the huge interdomain discrepancies and intradomain variances lead to complicated class-level relations. Specifically, the instances of the same class differ greatly, while instances of different classes are similar, whether across different RSIs domains or within the same RSIs domain. However, existing methods cannot fully consider these problems, limiting the performance of UDA semantic segmentation of RSIs. To this end, this article proposes a novel cross-domain multiprototypes learning method, the core idea of which is to abstract the cross-domain and intradomain class-level relations into multiple prototypes. Specifically, the multiple prototypes belonging to different classes can detailedly describe complex interclass relations, and the multiple prototypes within the same class can better model rich intraclass relations. Furthermore, the source and target samples are jointly used for prototypes calculation, to fully fuse the feature information of different RSIs. In a nutshell, utilizing the samples from different RSIs domains to learn multiple prototypes for each class can achieve better domain alignment at the class level. In addition, considering that RSIs simultaneously contain large targets with wide coverage and important small targets, two masked consistency learning strategies are designed to better explore the contextual structure of target RSIs and improve the quality of pseudo labels for prototype updating. The global consistency strategy can strengthen the utilization of global context relations, while the local consistency strategy can further improve the learning of local context details. Therefore, the proposed method is actually a prototype and context-enhanced learning method for UDA semantic segmentation of RSIs. Extensive experiments demonstrate that the proposed method can achieve better performance than existing state-of-the-art UDA methods.**

***Index Terms*— Context enhancement, cross-domain multiprototypes, remote sensing images (RSIs), semantic segmentation, unsupervised domain adaptation (UDA).**

I. Introduction

AIMING to adapt the model trained on a labeled source domain to an unlabeled target domain, the unsupervised domain adaptation (UDA) methods can significantly reduce the dependence on a large number of labeled target samples and effectively improve the adaptation ability of deep models [1], [2]. In recent years, the problem of UDA semantic segmentation of remote sensing images (RSIs) has attracted more and more attention, and many methods and explorations have emerged successively, achieving promising and instructive results [3], [4], [5]. However, according to the statistics, the performance of existing UDA methods for RSIs semantic segmentation is still far behind that of supervised counterparts, and there is still a lot of room for performance improvement.

As we all know, different RSIs have completely different resolutions, imaging conditions, and styles, which directly causes the large domain discrepancies. Even in the same RSI domain, different instances within the same class vary greatly in size, shape, color, and texture due to the wide coverage. Such interdomain discrepancies and intradomain variations inevitably lead to complicated class-level relations, whether across different RSIs domains or within the same RSIs domain. As shown in Fig. 1, on the one hand, the instances belonging to the same class are very distinct from each other, such as the building objects with completely different appearances and sizes. These distinctions manifest themselves simultaneously between instances within different domains and between instances within the same domain. On the other hand, the instances belonging to different classes are highly similar, such as tree and low vegetation. Similarly, these similarities also manifest themselves simultaneously between different classes within different domains and between different classes within the same domain. The above problems lead to the complicated class-level relations existing across different domains and within the same domain, which undoubtedly brings great challenges to RSIs domain adaptation. Therefore, how to better describe these complicated class-level relations is the key to further improve the performance of UDA semantic segmentation of RSIs.

Through drawing on the development of computer vision, many class-level alignment strategies have been introduced into RSIs domain adaptation [6]. For example, Chen et al. [7] embedded the class-certainty attention module into the widely used adversarial learning framework, and Xu et al. [8] designed a class-aware generative adversarial network (GAN) for UDA semantic segmentation of RSIs. However, according to the statistics, the improvement brought about by these methods is limited. More recently, in the field of computer vision, utilizing class prototypes to capture the class-level features in UDA tasks has been favored by more and more research [9]. For example, Yue et al. [10] unified prototype learning and self-supervised learning into a UDA framework. Xie et al. [11] proposed a prototype-based contrastive learning method. In addition, a variety of prototype-guided UDA techniques, such as prototypical pseudo label [12] and

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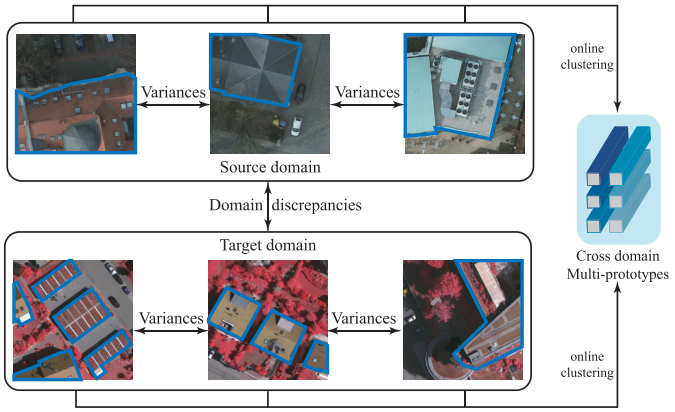


Fig. 1. Schematic of learning cross-domain multiple prototypes with the building class as an example.

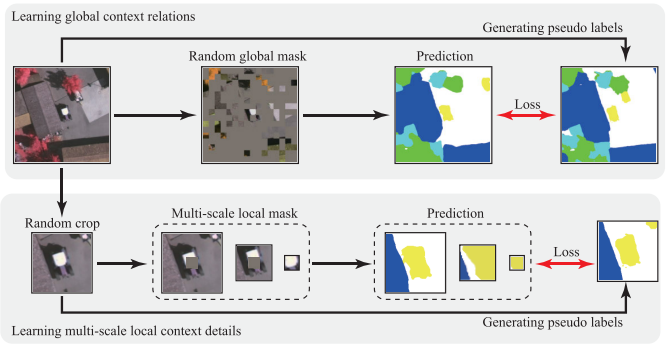


Fig. 2. Schematic of two masked consistency learning strategies. In the local consistency strategy, there are three input patches with different scales and the last one is a center crop without any mask.

prototype matching [13], have been proposed successively. Nevertheless, there are still several fundamental weaknesses in the above methods, which limit their application performance in UDA semantic segmentation of RSIs. First, these methods all learn just one prototype for each class. However, for the RSIs with large interdomain discrepancies and intradomain variances, this single-prototype learning mode is obviously insufficient to express complicated class-level relations. Second, the class information from different domains is not fully integrated and fused. For example, Yue et al. [10] estimated source and target prototypes through different memory banks and Wu et al. [13] calculated the patchwise prototypes only by using target features, which obviously weakens the interaction between source and target information and affects the effects of domain alignment. Third, the offline prototype calculation and multistep training pipeline are undoubtedly tedious, hindering the improvement of application potential and production efficiency.

In RSIs domain adaptation, the large interdomain discrepancies and intradomain variances can be boiled down to the complicated class-level relations. Therefore, we propose to learn the cross-domain multiprototypes for each class based on source and target RSIs, as shown in Fig. 1. The core idea is to abstract the cross-domain and intradomain class-level relations into multiple prototypes. Specifically, a set of prototypes are established for each class so that multiple prototypes corresponding to different classes can better describe rich interclass relations, while different prototypes within the same class can better model complex intraclass relations. Furthermore, these prototypes are calculated and updated jointly by source and target samples, to fully utilize and fuse the feature information from different domains. In addition, the online clustering strategy is adopted for prototypes calculation in the one-step training process, ensuring the coevolution of prototypes and the efficiency of RSIs domain adaptation training. In a nutshell, utilizing the samples from different RSIs domains to learn multiple prototypes for each class can achieve better domain alignment at the class level.

The calculation and updating of prototypes require both RSI samples and corresponding labels. Therefore, the self-supervised learning framework is adopted to provide pseudo labels for prototypes updating with unlabeled target samples. More importantly, considering that RSIs simultaneously contain large targets with wide coverage and important small targets, two masked consistency learning strategies are designed to better explore the contextual structure of target RSIs and improve the quality of pseudo labels. As shown in Fig. 2, the global consistency strategy can strengthen the utilization of global context relations by predicting the RSI with a random global mask, thus improving the segmentation performance of large targets with wide coverage. Also, the local consistency strategy can improve the learning of multiscale local context details by predicting multiple masked RSI patches with different spatial sizes, thus improving the segmentation accuracy of important small targets and regions. The common idea behind the two strategies is that the model must utilize unmasked regions to predict masked regions, thus effectively enhancing the utilization of global and local context information during RSIs domain adaptation. To sum up, the proposed method can improve RSIs domain adaptation from the two aspects of prototype enhancement and context enhancement. The main contributions include the following four points.

1) A prototype and context-enhanced learning method (PCEL) for UDA semantic segmentation of RSIs is proposed, to further improve the segmentation results of target RSIs.
2) A cross-domain multiprototypes learning method for RSIs domain adaptation is proposed. It can better model the complicated class-level relations and effectively fuse the feature information from different RSIs, thus further improving the effect of domain alignment.
3) Two masked consistency learning strategies are designed, to further improve the utilization of the global and local context information of target RSIs and the quality of pseudo labels for prototype updating.
4) Extensive experiments are conducted on airborne and spaceborne RSIs, and the results show that the performance of the proposed method is superior to that of existing state-of-the-art UDA methods.

II. Related Work

A. RSIs Semantic Segmentation

RSIs semantic segmentation, converting complex images into simple semantic maps through pixel-to-pixel classification, has become one of the most important approaches for RSIs intelligent interpretation [2]. In recent years, the

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continuous development of deep learning has given rise to a variety of methods with excellent segmentation performance, such as FCN [14], Unet [15], PSPNet [16], and UPerNet [17]. The introduction of these methods significantly boosts the performance of RSIs semantic segmentation [18], [19]. The encoder–decoder structure is the most widely used segmentation framework, and most of the existing methods are developed and improved on this basis [20], [21]. Furthermore, to compensate for the shortage that convolution kernels cannot make full use of long-distance dependencies, the attention mechanism has become popular [22], [23]. At the same time, the performance of the transformer model on RSIs semantic segmentation has also been widely explored [24], [25], [26], due to its excellent performance in natural image processing and analysis. In addition, the research on instance segmentation [27], [28], small object segmentation [29], and hyperspectral images segmentation [30], [31] has also been developed and paid more and more attention.

B. UDA Semantic Segmentation of RSIs

Due to the large discrepancies between different RSIs domains, the model well-trained on one dataset could hardly achieve satisfactory segmentation results on another dataset. To this end, the UDA methods are introduced to improve the adaptation ability of models to target RSIs. The existing methods could be roughly divided into three categories: RSI translation, adversarial learning, and self-supervised learning. Although the adopted learning modes and ideas are different, the above methods all attempt to improve the generalization performance of models on target RSIs by reducing domain discrepancies [32]. Drawing on the idea of image translation, many RSI translation methods have been designed. These methods can achieve pixel-level translation based on fully considering the RSIs characteristics, which is simple and effective. Tasar et al. [33] proposed a color mapping GAN for UDA semantic segmentation of RSIs. Zhao et al. [34] designed two novel methods resize-residual DualGAN (RDG) and depth-assisted resize-residual DualGAN (DRDG) by introducing the in-network resizer module and residual connection into the widely used DualGAN model [35]. Meanwhile, the performance of adversarial-based UDA methods on RSIs domain adaptation has also been widely studied. Yan et al. [36] proposed a novel triplet adversarial network to learn a domain invariant classifier by jointly considering both source and target domains. Ma et al. [37] introduced a mutually boosted attention module into the GAN network and aligned global feature maps while enhancing domain general features. Zheng et al. [38] improved the structure of HRNet based on the characteristics of RSIs data and embedded it into the general adversarial learning framework. However, it is undeniable that the process of adversarial training is a very tricky issue due to the instability and the difficulty of convergence [39]. In contrast, the self-supervised-based UDA methods can effectively improve the performance of RSIs domain adaptation by fully utilizing pseudo labels, and its training process is more stable and the obtained results are more robust. Wang et al. [1] focused on the problem of spatial resolution inconsistency in RSIs domain adaptation, effectively improving the effect of knowledge transfer from airborne to spaceborne images. Li et al. [40] introduced the gradual class weights strategy and local dynamic quality strategy into the SegFormer model and constructed a self-supervised framework for UDA semantic segmentation of RSIs. Yan et al. [41] designed a cross teacher–student network and improved the classification results on target RSIs through the cross consistency training.

More recently, some attempts to combine these different types of methods have begun to emerge, further improving the performance of RSIs domain adaptation. For example, Liang et al. [42] developed a multilevel UDA method based on the instance-level style transfer technique, the feature-level contrastive learning, and the decision-level task consistency training. Bai et al. [43] designed an integrated method of contrastive learning and adversarial learning for RSIs domain adaptation. Cai et al. [44] introduced the super-resolution task into UDA semantic segmentation and present a novel multitask network. In additional, Li et al. [45] investigated the performance of different GAN models on RSI unsupervised style transfer.

Although various UDA methods for RSIs semantic segmentation have been proposed, they do not explicitly explore the complicated class-level relations, and there is still a lot of room for performance improvement. Based on the self-supervised learning framework, the proposed method can effectively enhance the utilization of prototypical features and context information, to further improve the performance of UDA semantic segmentation of RSIs.

C. Prototype-Guided UDA

As an abstract embedding of essential semantic representations, class prototypes indeed play an active role in enhancing intraclass compactness and interclass separability. Recently, in the field of computer vision, more and more research works begin to introduce prototype learning into domain adaptation training [9], [10], [11], [12], [13], to better capture and exploit class-level features. Wu et al. [13] proposed a patchwise prototypical matching method combined with style transfer for UDA semantic segmentation. Zhang et al. [12] relied on representative prototypes to denoise pseudo labels and gather target features in the process of self-supervised domain adaptation. Lu et al. [46] established the anisotropic prototypes to represent each class. Jiang et al. [47] proposed a prototypical contrast adaptation method to conduct class-centered distribution alignment. Zhu et al. [48] designed an invariant feature memory module to store invariant domain-level context information. However, as mentioned above, these methods tailored to street scene segmentation generally adopt the single-prototype learning mode and hinder the information interaction between source and target domains, which greatly limits their application performance in UDA semantic segmentation of RSIs.

In the field of remote sensing, to the best of our knowledge, this article is the first to utilize prototype learning to improve the performance of UDA semantic segmentation of RSIs. Different from the above prototype-guided UDA methods, the proposed cross-domain multiprototypes learning method aims to abstract complicated class-level relations into multiple prototypes, so as to effectively alleviate the difficulties brought

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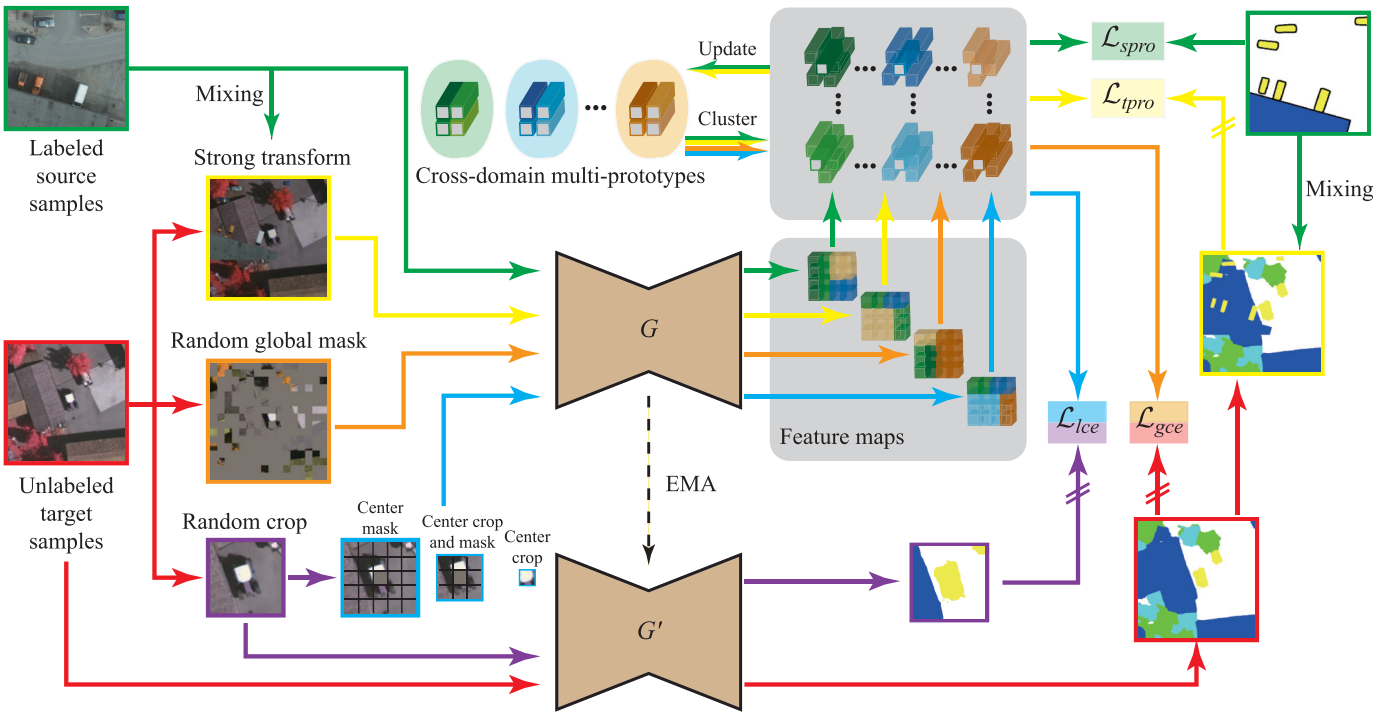


Fig. 3. Workflow of the proposed method. G and G' denote the student model and the teacher model generated by the exponential moving average (EMA) algorithm. The model G receives four different data flows, which are source samples, target samples after strong transform and mixing, target samples with the random global mask, and target samples with different local operations from top to bottom. The feature maps generated by G are clustered according to current prototypes, and in turn, the cluster results are used for prototype updating and label inference. The model G' provides pseudo labels for prototype updating and masked consistency learning. The losses $\mathcal{L}_{\text{spro}}$ and $\mathcal{L}_{\text{tpro}}$ are used for cross-domain multiprototypes learning, while the losses $\mathcal{L}_{\text{gce}}$ and $\mathcal{L}_{\text{lce}}$ are used for masked consistency learning.

by interdomain discrepancies and intradomain variations for RSIs domain adaptation. In addition, two masked consistency strategies are designed to further improve the quality of pseudo labels for prototype updating.

D. Context-Enhanced UDA

In the field of computer vision, more and more research works begin to attempt to further enhance the performance of UDA by fully exploring the high-quality target domain information. Most of the methods are based on self-supervised training and contrast learning. For example, Xie et al. [49] proposed a semantic-guided pixel contrast method, to focus on the intrinsic connections among pixels with similar semantic concepts. Li et al. [50] effectively improved the exploration and utilization of target domain information from the views of boundaries and features. Hoyer et al. [51] applied a novel consistency learning module into UDA semantic segmentation. Chen et al. [52] developed a unified pixelwise and patchwise self-supervised learning method to improve the learning of target context information.

The above research works fully verify the feasibility and effectiveness of exploring context information in target domains. Inspired by these research works but clearly different from them, two masked consistency learning strategies are designed to further improve the performance of RSIs domain adaptation by fully exploring the contextual structure of target RSIs. More importantly, our method fully considers the data characteristics of RSIs. Specifically, for the large objects with wide coverage in RSIs, the global consistency strategy is designed to enhance the capture of global context relations, and for the small important targets and regions, the local consistency strategy is designed to enhance the learning of local context details at different scales.

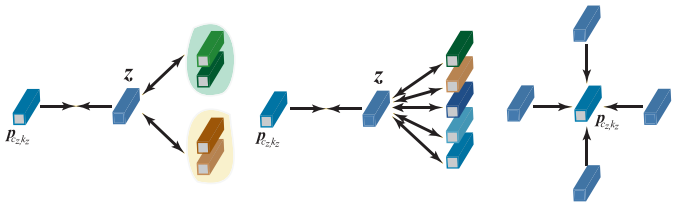
III. Methodology

A. Workflow

To further improve the performance of UDA semantic segmentation of RSIs, a novel method PCEL is proposed from the two aspects of prototype enhancement and context enhancement. Fig. 3 shows the workflow of the proposed method, which is developed based on the efficient self-supervised learning framework. Specifically, the proposed method mainly consists of four different data flows.

The source samples are first mapped into feature vectors by the student model G, which are then clustered according to current prototypes. The obtained cluster results directly determine the prediction class of each pixel in the input samples, and then, the prediction is used for loss calculation with true labels. Similarly, the target samples after strong transform and sample mixing are also used for clustering and loss calculation with the pseudo labels generated by the teacher model G', to complete the adaptation to target RSIs. During the above processes, two important steps of online clustering and prototype updating are performed for cross-domain multiprototype learning. The former refers to assigning feature vectors to different prototypes of different classes, while the latter is to update prototypes according to this assignment. Learning multiple prototypes for each class can better model the complex intraclass and interclass relations, while the cooperative participation of source and

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(a) (b) (c)

Fig. 4. Illustrations of class-level relations optimization. (a) Interclass relation. (b) Intercluster relation. (c) Intracluster compactness.

target samples can make the learned prototypes contain the feature information from different RSIs and further improve the effect of domain alignment.

Considering that RSIs simultaneously contain targets of different sizes, the two data flows of target samples with different masked consistency strategies are added, to better explore the contextual structure of target RSIs and improve the quality of pseudo labels. In the global consistency strategy, the whole sample is first masked randomly, and then, its prediction is compared with the pseudo label of the sample without mask, to enhance the learning of global context relations. In the local consistency strategy, several small patches are first obtained randomly. For each patch, three different operations, including center mask, center crop and mask, and center crop, are performed to further generate three patches with different masks and spatial sizes. Predicting these masked patches can effectively enhance the learning of local context details at different scales. The two strategies can effectively facilitate the capture of global and local context information, to further improve the effect of multiprototype-based domain adaptation. Note that only clustering and label prediction are performed in the above processes, without prototype updating.

Next, the cross-domain multiprototype learning and masked consistency learning strategies are first described in detail, and then, the overall optimization objective and the pseudocode of workflow are given.

B. Cross-Domain Multiprototypes

1) Class-Level Relations Optimization: The purpose of learning cross-domain multiprototypes is to better model the complicated class-level relations in RSIs domain adaptation, which can be summarized into three aspects. First, the prototypes of different classes should be separated from each other, to better distinguish the classes containing lots of similar instances. Furthermore, for the significantly different instances of the same class, different prototypes should be established and appropriately distanced, to describe the rich intraclass relations. In addition, the compactness of each prototype-centered cluster should be enhanced, to further improve the discrimination between different prototypes. Next, the three loss functions designed for the above purposes will be introduced in turn.

Formally, the input source or target RSI sample is denoted as $\boldsymbol{I} \in \mathcal{R}^{h\times w\times 3}$, and the model G first maps it to the feature map $\boldsymbol{Z} \in \mathcal{R}^{H\times W\times D}$, which can also be denoted as $\boldsymbol{Z} = \{z_n\}_{n=1}^{N}$ $(N = H \times W)$. For the segmentation task with C classes, the prototypes are denoted as $\boldsymbol{P} = \{\boldsymbol{p}_{c,k} \in \mathcal{R}^D\}_{c,k=1}^{C,K}$, in which K is the number of prototypes per class and each prototype is determined as the center of kth subcluster of features belonging to the same class. In fact, each feature vector z in the feature map $\boldsymbol{Z}$ corresponds to a different pixel in the input RSI sample. Therefore, the label of each pixel can be inferred through nearest prototype retrieving [53] between z and $\boldsymbol{p}_{c,k}$

$$\hat{c}_z = c^*, \text{with}(c^*, k^*) = \underset{(c,k)}{\text{argmin}}\{\langle z, \boldsymbol{p}_{c,k}\rangle\}_{c,k=1}^{C,K} \tag{1}$$

where the distance function $\langle\cdot,\cdot\rangle$ is defined as the negative cosine similarity. Based on the inference obtained by (1) and the true label (pseudo label for target sample) $c_z \in \{1,\ldots,C\}$, the cross-entropy loss can be calculated

$$\mathcal{L}_{\text{ce}} = -\log p(c_z \mid z) = -\log\frac{\exp(-s_{z,c_z})}{\exp(-s_{z,c_z}) + \sum_{c'\neq c_z}\exp(-s_{z,c'})} \tag{2}$$

where $p(c \mid z)$ is the probability distribution of the feature z over C classes and $s_{z,c} = \min\{\langle z, \boldsymbol{p}_{c,k}\rangle\}_{k=1}^{K}$ is the distance between the feature and the closet prototype of class c. As shown in Fig. 4(a), (2) can effectively model the interclass relation by reducing the distance between the feature and the nearest prototype and increasing the distance between it and other prototypes of irrelevant classes, to improve the separability between different classes.

As shown in Fig. 4(b), within the same class, the feature should be pushed further close to its corresponding prototype and relatively away from other prototypes, to better describe the rich intraclass relations. Therefore, the pixel-prototype contrastive loss is calculated by maximizing the prototype assignment posterior probability

$$\mathcal{L}_{\text{ppc}} = -\log\frac{\exp(z^\top \boldsymbol{p}_{c_z,k_z}/\tau)}{\exp(z^\top \boldsymbol{p}_{c_z,k_z}/\tau) + \sum_{\boldsymbol{p}^-\in\mathcal{P}^-}\exp(z^\top \boldsymbol{p}^-/\tau)}$$
$$\text{s.t. } k_z = \text{argmax}_k\{l_{z,k}\}_{k=1}^{K},\ l_{i,k} \in \boldsymbol{l}_z \tag{3}$$

where $\mathcal{P}^- = \{\boldsymbol{p}_{c,k}\}_{c,k=1}^{C,K}/\boldsymbol{p}_{c_z,k_z}$, the hyperparameter τ is used for controlling the concentration level, and k_z indexes the prototypes belonging to class c. Intuitively, (3) can reduce the distance between each feature and its assigned prototype $\boldsymbol{p}_{c_z,k_z}$ (positive prototype) and increase the distance between each feature and other $CK-1$ prototypes $\mathcal{P}^-$ (negative prototypes) belonging to the same class.

To further improve the compactness and discriminability of each prototype-centered cluster, the features should be more tightly distributed around the corresponding prototype, which can be implemented by the pixel-to-prototype distance loss

$$\mathcal{L}_{\text{ppd}} = (1 - z^\top \boldsymbol{p}_{c_z,k_z})^2 \tag{4}$$

where both the z and $\boldsymbol{p}_{c_z,k_z}$ are $L2$ normalized. As shown in Fig. 4(c), (4) can minimize the distance between each feature and its assigned prototype, effectively reducing intracluster variances.

At this point, in the process of cross-domain multiprototypes learning, the total loss can be given

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_1\mathcal{L}_{\text{ppc}} + \lambda_2\mathcal{L}_{\text{ppd}} \tag{5}$$

where λ_1 and λ_2 are weight coefficients. Note that the loss $\mathcal{L}$ can be specifically denoted as $\mathcal{L}_{\text{spro}}$ and $\mathcal{L}_{\text{tpro}}$ when the

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source and target samples are used for prototypes calculation, respectively. The losses $\mathcal{L}_{\text{ce}}$, $\mathcal{L}_{\text{ppc}}$, and $\mathcal{L}_{\text{ppd}}$ can optimize the learning process from three different levels of interclass relation, intercluster relation, and intracluster compactness, respectively. Consequently, clustering different features around different prototypes can better model interclass and intraclass relations, which is very important and effective for RSIs domain adaptation.

2) Online Clustering for Label Prediction: Given the features $\boldsymbol{Z}^c$ belonging to the class c in a batch, the goal of online clustering is to assign the features $\boldsymbol{Z}^c$ to the K prototypes of the class c, which actually completes the inference on the class labels of pixels. This assignment process can be denoted as $\boldsymbol{L}^c = [\boldsymbol{l}_{z_n}]_{n=1}^N$, where $\boldsymbol{l}_{z_n} = [l_{z_n,k}]_{k=1}^K \in \{0,1\}^K$ is the one-hot vector assigning features z_n to the K prototypes. The probability matrix $\boldsymbol{L}^c$ can be optimized by maximizing the similarity between the features $\boldsymbol{Z}^c$ and prototypes $\boldsymbol{P}^c$

$$\begin{aligned} &\max_{L^c} \operatorname{Tr}(\boldsymbol{L}^{c\top} \boldsymbol{P}^{c\top} \boldsymbol{Z}^c) \\ &\text{s.t. } \boldsymbol{L}^c \in \{0,1\}^{K\times N}, \quad \boldsymbol{L}^{c\top}\mathbf{1}^K = \mathbf{1}^N,\ \boldsymbol{L}^c\mathbf{1}^N = \frac{N}{K}\mathbf{1}^K \end{aligned} \tag{6}$$

where $\mathbf{1}^N$ is the vector with N-dimensional elements that are all ones. The constraint $\boldsymbol{L}^{c\top}\mathbf{1}^K = \mathbf{1}^N$ ensures that each feature can only be assigned to one prototype. The constraint $\boldsymbol{L}^c\mathbf{1}^N = (N/K)\mathbf{1}^K$ ensures that each prototype can be selected at least N/K times in a batch on average, which can effectively prevent the model from being optimized to extreme and trivial solutions, such as assigning all features to a single prototype. The solution of (6) can be completed by relaxing $\boldsymbol{L}^c$ to be an element of the transportation polytope

$$\begin{aligned} &\max_{L^c} \operatorname{Tr}(\boldsymbol{L}^{c\top} \boldsymbol{P}^{c\top} \boldsymbol{Z}^c) + \kappa h(\boldsymbol{L}^c) \\ &\text{s.t.} \quad \boldsymbol{L}^c \in \mathcal{R}_+^{K\times N},\ \boldsymbol{L}^{c\top}\mathbf{1}^K = \mathbf{1}^N,\ \boldsymbol{L}^c\mathbf{1}^N = \frac{N}{K}\mathbf{1}^K \end{aligned} \tag{7}$$

where $\kappa > 0$ is the hyperparameter controlling the smoothness of distribution and $h(\boldsymbol{L}^c) = \sum_{n,k} -l_{z_n,k}\log l_{z_n,k}$ is an entropy. Furthermore, the solver of (7) can be given based on the soft assignment relaxation and the regularization term $h(\boldsymbol{L}^c)$

$$\boldsymbol{L}^c = \operatorname{diag}(\boldsymbol{u})\exp\left(\frac{\boldsymbol{P}^{c\top}\boldsymbol{Z}^c}{\kappa}\right)\operatorname{diag}(\boldsymbol{v}) \tag{8}$$

where $\boldsymbol{u} \in \mathcal{R}^K$ and $\boldsymbol{v} \in \mathcal{R}^N$ are renormalization vectors computed by few steps of Sinkhorn–Knopp iteration [54].

3) Prototype Updating With Source and Target RSI Samples: To make the learned prototypes contain the feature information from different RSIs domains, the source and target samples are jointly used for prototypes calculation and updating. As described above, the prototypes are constructed directly on input samples themselves, without introducing additional parameters that need to be optimized. Specifically, the prototypes are actually calculated as the centers of the corresponding features. Therefore, the prototypes can be continuously updated by taking full account of the online clustering results

$$\boldsymbol{p}_{c,k} \leftarrow \mu \boldsymbol{p}_{c,k} + (1-\mu)\bar{\boldsymbol{z}}_{c,k} \tag{9}$$

where $\mu \in [0,1]$ is the coefficient for momentum updating and $\bar{\boldsymbol{z}}_{c,k}$ represents the mean vector of features after $L2$ normalization. It should be noted that (9) applies to both source and target samples in a batch, and the learned prototypes that have fused the rich feature information from different RSIs domains can prompt the better domain alignment at the class level.

C. Masked Consistency Learning

On the one hand, RSIs simultaneously contain large targets with wide coverage and important small targets, which forms the main contextual structure of observation areas. On the other hand, the generative self-supervised training based on masked images has been proved to be an effective way to learn rich contextual features in the field of computer vision [51], [55]. Therefore, drawing from the idea of masked training, two masked consistency strategies are designed to fully learn the global and local context information in target RSIs and further improve the quality of pseudo labels in the process of domain adaptation training.

1) Global Masked Consistency: For each target RSI sample $i_t \in \mathcal{R}^{h\times w\times 3}$, a global mask M is first randomly generated

$$M_{h_s b+1:(h_s+1)b, w_s b+1:(w_s+1)b} = [v > r] \quad \text{with } v \in \mathcal{U}(0,1) \tag{10}$$

where $\mathcal{U}$ is the uniform distribution, $[\cdot]$ is the Iverson bracket, and b is the spatial size of a masked patch. In addition, r is the mask ratio, and $h_s \in [0,\ldots,h/b-1]$ and $w_s \in [0,\ldots,w/b-1]$ index the patch. Then, the masked sample is obtained by elementwise multiplication, i.e., $i_t^M = M \odot i_t^T$, and its prediction by (1) can be denoted as $\hat{y}^M$. Furthermore, the global consistency loss is calculated as

$$\mathcal{L}_{\text{gce}} = q^T \mathcal{H}\left(\hat{y}^M, \tilde{y}^M\right) \tag{11}$$

where $\mathcal{H}$ denotes the cross-entropy function, q^T is the quality weight, and $\tilde{y}^M$ is the pseudo label of the original sample i_t. According to (10) and (11), the model has to utilize the remaining context clues in the masked sample to reconstruct the correct label, so as to enhance the learning of global context relations.

2) Local Masked Consistency: As described above, the local consistency learning consists of three prediction processes at different scales. Specifically, for each target RSI sample, several patches with the size of $5b \times 5b$ is randomly cropped, which can be denoted as i_t^{cp} Then, three different operations are performed: 1) mask out the center region with the size of $b \times b$; 2) center crop with the size of $3b \times 3b$ and mask out the center region with the size of $b \times b$; and 3) center crop with the size of $b \times b$ and no mask. The generated results can be denoted as i_t^{cp5}, i_t^{cp3}, and i_t^{cp1}, and Fig. 3 shows these operations vividly. The first two are designed to force the model to reconstruct the correct labels of the center region by using its surrounding context information, while the last one is to directly predict labels of small center regions. Similarly, the predictions of i_t^{cp5}, i_t^{cp3}, and i_t^{cp1}, and the pseudo labels of i_t^{cp} are used for loss calculation with reference to (11). The total loss of local consistency learning is the average value of these losses, which can be denoted as

$$\mathcal{L}_{\text{lce}} = (\mathcal{L}_{\text{lce5}} + \mathcal{L}_{\text{lce3}} + \mathcal{L}_{\text{lce1}})/3 \tag{12}$$

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where $\mathcal{L}_{\text{lce5}}$, $\mathcal{L}_{\text{lce3}}$, and $\mathcal{L}_{\text{lce1}}$ denote the losses of the three prediction processes. Obviously, (12) can effectively prompt the learning of local context details at different scales.

Undoubtedly, the random mask strategy can force the model to predict the masked regions only using the remaining unmasked regions. The global masked consistency can enhance the utilization of global context relations and thus improve the segmentation performance of large targets with wide coverage. The local masked consistency can prompt the learning of local context details and thus improve the segmentation accuracy of important small targets and regions. Consequently, through fully exploring the contextual structure of target RSIs, the two masked consistency strategies can further improve the effect of self-supervised domain adaptation and the quality of pseudo labels for prototype updating.

D. Optimization Objective

As shown in Fig. 3, the proposed PCEL consists of four losses, i.e., $\mathcal{L}_{\text{spro}}$, $\mathcal{L}_{\text{tpro}}$, $\mathcal{L}_{\text{gce}}$, and $\mathcal{L}_{\text{lce}}$. The first two are used for cross-domain multiprototypes learning, while the second two are used for masked consistency learning. Therefore, the total loss can be calculated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{spro}} + \mathcal{L}_{\text{tpro}} + \gamma(\mathcal{L}_{\text{gce}} + \mathcal{L}_{\text{lce}}) \tag{13}$$

where γ is used to adjust the proportion of the two masked consistency losses. By optimizing (13), both prototype enhancement and context enhancement can be achieved in the proposed PCEL, to further improve the performance of UDA semantic segmentation of RSIs. In addition, Algorithm 1 summarizes the pseudocode to clearly show the entire workflow.

Algorithm 1 PCEL for UDA Semantic Segmentation of RSIs

Require: I_t, I_s : unlabeled target sample and labeled source sample
Require: G, G' : student model and teacher model
Require: $\boldsymbol{P}$: cross-domain multi-prototypes
randomly initialize G, G' and $\boldsymbol{P}$
while not done **do**
  Sample batch (i_t, i_s) from (I_t, I_s)
  for (i_t, i_s) **do**
    Calculate $\mathcal{L}_{spro}$ with Equation 5
    Calculate $\mathcal{L}_{tpro}$ with Equation 5
    Update $\boldsymbol{P}$ with Equation 9
  end for
  for i_t **do**
    Generate i_t^M, i_t^{cp5}, i_t^{cp3} and i_t^{cp1} with masks
    Calculate $\mathcal{L}_{gce}$ and $\mathcal{L}_{lce}$ with Equations 11 and 12
  end for
  Calculate $\mathcal{L}_{total}$ with Equation 13
  Update G and G'
end while

IV. Experimental Results

A. Datasets Description

The two widely used RSIs datasets ISPRS and LoveDA [56] are used for experiments. The ISPRS dataset contains two subsets Vaihingen (VH) and Potsdam (PD), and the LoveDA dataset also contains two subsets Rural and Urban. The details are listed in Table I.

TABLE I
Details of Different RSIs Datasets

Datasets	Subsets	Types	Coverage	Resolution	Bands
ISPRS	VH	Airborne	1.38 km^2	0.09 m	IRRG
ISPRS	PD	Airborne	3.42 km^2	0.05 m	RGB IRRG RGBIR
LoveDA	Rural Urban	Spaceborne	536.15 km^2	0.3 m	RGB

The VH dataset contains 33 images, which are divided into two groups consisting of 16 and 17 images according to the official statements. The PD dataset consists of 38 images, which are divided into two groups consisting of 24 and 14 images. To further increase the domain discrepancies, the PD dataset with the RGB imaging mode is adopted. Both the two datasets contain the same six classes: impervious surface, building, low vegetation, tree, car, and cluster. During preprocessing, all images are cropped to samples with 512×512 pixels. Consequently, the VH dataset produces 344 and 398 samples, while the PD dataset produces 3456 and 2016 samples.

Compared with the ISPRS datasets, LoveDA is a more challenging spaceborne RSIs dataset. The Rural dataset contains 1366 training images and 992 validation images, and the Urban dataset contains 1156 training images and 677 validation images. Each image is 1024×1024 pixels. Both the two datasets contain seven classes, including background, building, road, water, barren, forest, and agricultural.

Referring to [5] and [57], the three UDA tasks, PD→VH, VH→PD, and Rural→Urban, are designed. In the first task, the 3456 labeled samples of PD and the 398 unlabeled samples of VH are used for domain adaptation training, while the labeled 344 samples of VH are used for evaluating. In the second task, the 344 labeled samples of VH and the 2016 unlabeled samples of PD are used for domain adaptation training, while the labeled 3456 samples of PD are used for evaluating. In the third task, the 1366 labeled samples of Rural and the 677 unlabeled samples of Urban are used for domain adaptation training, while the labeled 1156 samples of Urban are used for evaluating.

B. Experimental Settings

The hardware environment includes a computer equipped with an Intel Xeon Gold 6152 CPU and an Nvidia A100 PCIE GPU. The software environment is developed on the ubuntu system, including Python 3.9 and machine learning libraries.

Referring to [40], [58], and [59], the advanced segmentation framework SegFormer is adopted. Specifically, the standard Mix Transformer-B5 model pretrained on the ImageNet dataset is used as the backbone network since the transformer models have achieved excellent performance in RSIs semantic segmentation. The decoder network first utilizes the 1×1 convolution and bilinear upsampling for channel and size alignment of the outputs at different levels of the backbone and then integrates multiscale feature information with the improved atrous spatial pyramid pool (ASPP) module.

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Different from the standard settings, the dimension of the output features of the improved ASPP module is variable and is actually equal to the dimension of prototypes.

During domain adaptation training, the AdamW algorithm is used for parameter optimization, where the parameters β_1 and β_2 are set to 0.9 and 0.999, respectively, and the weight decay coefficient is set to 0.01. The initial learning rates of the backbone and decoder network are set to 6×10^{-5} and 6×10^{-4}, respectively, to obtain faster convergence. The max iteration is set to 40 000. Before 1500 iterations, the linear warm-up strategy is adopted, and after that, the learning rate decays linearly. The same sample augmentation and mixing strategies as in [60] are applied, to improve the robustness of self-supervised training and the generalization ability of the trained model. To ensure the quality of pseudo labels, the pixels with the prediction probability larger than 0.968 are used for loss calculation and prototypes updating. The coefficient μ for prototypes updating is set to 0.999. The weight coefficients λ_1 and λ_2 are set to 0.01 and 0.001, respectively, and the hyperparameter γ is set to 1. The batch size of each RSIs domain is set to 8, which means that each training batch contains eight labeled source samples and eight unlabeled target samples. For the PD→VH and VH→PD tasks, the number of prototypes per class is set to 10 and the dimension of prototype is set to 128. For the Rural→Urban task, the number of prototypes per class is set to 20 and the dimension of prototype is set to 256, considering that spaceborne RSIs have greater interdomain discrepancies and intradomain variations. The spatial size of masked patches and mask ratio is set to 32 and 0.7, respectively. In addition, during each iteration, four patches are randomly generated for local masked consistency learning.

Referring to [8], [61], and [4], the pixel accuracy (PA), intersection over union (IoU) per class, and mean value of IoU (mIoU) are used to quantitatively evaluate the segmentation performance of different methods on target RSIs.

C. UDA Methods for Comparison

To verify the effectiveness of the proposed method in UDA semantic segmentation of RSIs, eight advanced methods are used for comparison, which is listed as follows.

1) *Advent [62]:* A domain adaptation method combining the entropy minimization of pixelwise predictions and adversarial training.
2) *Zheng's [5]:* An entropy-guided adversarial learning method with local feature alignment and graph convolution for UDA semantic segmentation of RSIs.
3) *RDG [34]:* An RSI translation method focusing on the problem of scale discrepancy and translation stability between different RSI domains.
4) *DRDG [35]:* An RSI translation method combining depth information, which can further reduce the domain discrepancies between synthetic and target images. Due to the lack of digital surface model (DSM) data, this method is not applied to the LoveDA dataset.
5) *ProCST [63]:* An advanced image translation method based on a multiscale neural network. Referring to the original literature, this method will be combined with other methods such as self-supervised learning for domain adaptation segmentation.
6) *SAC [64]:* A self-supervised learning method can improve the effect of domain adaptation with the consistency training of the semantic predictions across multiscale augmentations and fusions.
7) *ProDA [12]:* An advanced self-supervised learning method aiming to denoise pseudo labels with representative prototypes and compact target feature spaces with prototype alignment. Experiments show that this method can effectively improve the effect of adaptation and transfer to target domains.
8) *CPSL [65]:* An advanced self-supervised domain adaptation method can effectively optimize self-training process by simultaneously rectifying pseudo labels with cluster assignments and improving the performance over long-tailed categories with class distribution alignment.
9) *Li's [40]:* An advanced transformer-based UDA method for RSIs semantic segmentation. This method introduces the gradual class weights strategy and the local dynamic quality strategy into the self-supervised learning process, effectively addressing the problem of class imbalance and improving the quality of pseudo labels.
10) *DAFormer [59]:* An advanced self-supervised learning method can significantly improve the performance of UDA semantic segmentation. Specifically, the proposed method utilizes the advanced transformer model as the backbone network for feature learning and utilizes three strategies of rare class sampling, feature distance calculation, and learning rate warm-up, to further improve the segmentation accuracy of target domains. In addition, the performance of this method combined with ProCST (denoted as DA-ProCST) is also used for comparison.

D. Results of ISPRS Dataset

The segmentation results of different methods on the ISPRS dataset are listed in Tables II and III, from which four observations can be obtained. First, the segmentation performance of the two adversarial learning methods, Advent and Zheng's, is extremely limited. This indicates that only utilizing the adversarial training process to align global feature distributions could hardly generate satisfactory results because neglecting class information will inevitably lead to misalignment and poor generalization. The results of the RSI translation method RDG are generally comparable to those of Zheng's, and the performance of DRDG is slightly better than Zheng's. Obviously, there is also much room for improvement. Second, the three self-supervised learning methods, SAC, ProDA, and CPSL, can utilize different strategies to optimize the self-training process, bringing about great improvements in the segmentation results of target RSIs. Compared to SAC, the methods ProDA and CPSL can make better use of the class-level information with representative prototypes and class distribution alignment and thus produce the higher mIoU and PA. Third, the two advanced domain adaptation methods Li's and DAFormer can significantly improve the segmentation results. Their common advantages lie in the adoption of the transformer model and class sampling strategy, which can effectively improve the adaptation and transfer to target

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TABLE II

SEGMENTATION PERFORMANCE OF DIFFERENT METHODS IN THE PD→VH TASK. IOU PER CLASS IS LISTED FROM COLUMN 3 TO COLUMN 8. BOLD VALUES INDICATE THE BEST RESULTS

Types	Methods	Venue	Impervious surface	Building	Low vegetation	Tree	Car	Clutter	mIoU	PA
Adversarial learning	Advent	CVPR 2019	55.43	68.49	20.73	59.02	28.28	0.80	38.79	65.51
	Zheng's	TGRS 2022	55.06	72.73	31.54	55.40	21.73	3.89	40.06	67.50
RSI translation	RDG	RS 2023	53.88	74.22	22.52	58.11	29.89	3.98	40.43	66.44
	DRDG	GRSL 2023	55.73	75.08	21.34	60.02	33.39	4.07	41.61	69.23
Self-supervised learning	SAC	CVPR 2021	69.06	79.78	29.14	66.73	50.34	10.19	50.87	73.88
	ProDA	CVPR 2021	77.26	82.64	50.12	68.92	47.37	8.21	55.76	81.42
	CPSL	CVPR 2022	73.92	84.19	44.68	72.18	60.28	20.69	59.32	81.23
	Li's	RS 2022	**80.27**	82.69	56.34	74.21	43.03	40.08	62.77	84.82
	DAFormer	CVPR 2022	77.76	89.29	53.81	74.08	65.04	33.82	65.63	84.93
	DA-ProCST	arXiv 2022	78.39	90.36	53.17	75.36	65.89	34.51	66.28	85.17
	PCEL		75.80	**93.12**	**58.28**	**78.42**	**71.32**	**63.69**	**73.44**	**86.60**

TABLE III

SEGMENTATION PERFORMANCE OF DIFFERENT METHODS IN THE VH→PD TASK. IOU PER CLASS IS LISTED FROM COLUMN 3 TO COLUMN 8. BOLD VALUES INDICATE THE BEST RESULTS

Types	Methods	Venue	Impervious surface	Building	Low vegetation	Tree	Car	Clutter	mIoU	PA
Adversarial learning	Advent	CVPR 2019	49.80	54.85	40.19	26.94	46.71	0.20	36.45	60.03
	Zheng's	TGRS 2022	47.63	48.77	34.92	41.17	51.58	0.79	37.48	60.89
RSI translation	RDG	RS 2023	52.17	48.00	40.01	37.69	44.47	1.42	37.29	60.63
	DRDG	GRSL 2023	54.05	50.53	39.14	39.15	47.08	1.36	38.55	62.54
Self-supervised learning	SAC	CVPR 2021	70.55	77.75	34.56	48.64	53.21	5.77	48.41	71.81
	ProDA	CVPR 2021	67.67	78.59	47.01	45.02	72.20	2.76	52.21	74.59
	CPSL	CVPR 2022	58.35	75.60	60.23	42.47	77.47	15.46	54.93	74.90
	Li's	RS 2022	**72.64**	82.39	54.12	48.69	60.51	18.05	56.07	77.98
	DAFormer	CVPR 2022	66.09	78.23	63.06	56.83	**77.57**	21.47	60.54	79.25
	DA-ProCST	arXiv 2022	66.86	79.31	62.76	59.20	77.10	20.81	61.01	79.73
	PCEL		65.04	**82.64**	**63.09**	**70.96**	76.95	**21.53**	**63.37**	**81.32**

RSIs domains. In addition, the introduction of the advanced translation method ProCST can further improve the domain adaptation performance of DAFormer. Fourth, the proposed method can achieve the best segmentation performance in the two UDA tasks. Specifically, compared to the second place, the mIoU of PCEL increases by 7.16% and 2.56% in the VH and PD datasets, respectively, and the PA increases by 1.43% and 1.59%, respectively. On the one hand, the proposed method can better describe complex intraclass and interclass relations through cross-domain multiprototypes learning, to comprehensively improve the recognition accuracy of different classes. For the classes that are relatively poorly distinguished by other methods, such as clutter in the VH dataset and tree in the PD dataset, the improvements resulting from the proposed method are significant. On the other hand, the designed consistency learning strategies can promote the utilization of global and local context information, further improving the segmentation performance of target RSIs. The global consistency strategy can effectively improve the segmentation accuracy of objects with large coverage, such as building in the VH and PD datasets, while the local consistency strategy can further optimize the recognition results of small-scale targets, such as car in the VH dataset and tree in the PD dataset.

In addition to the above quantitative statistics, several representative segmentation results are visualized for qualitative comparison and analysis, as shown in Figs. 5 and 6. It can be seen that there are a lot of misrecognition and class confusion in the segmentation maps of ProDA and CPSL, which are particularly prominent in the fourth line of Fig. 5(c) and the last

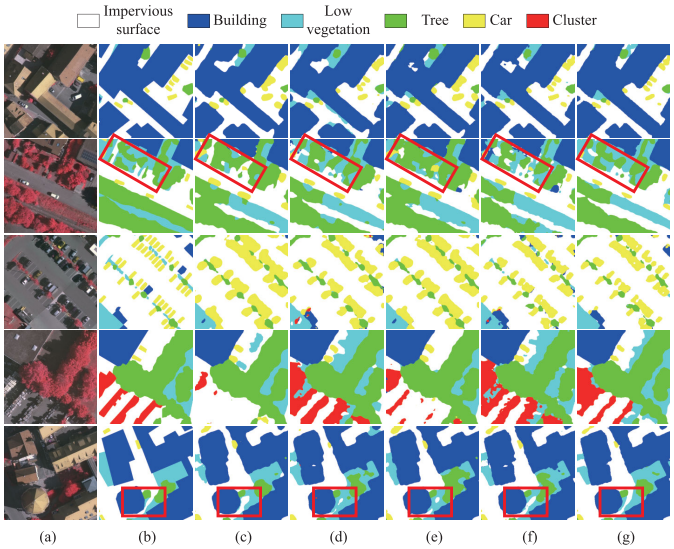


Fig. 5. Segmentation maps of different methods in the PD→VH task. (a) RSIs. (b) Ground truth. (c) ProDA. (d) CPSL. (e) Li's. (f) DAFormer. (g) PCEL.

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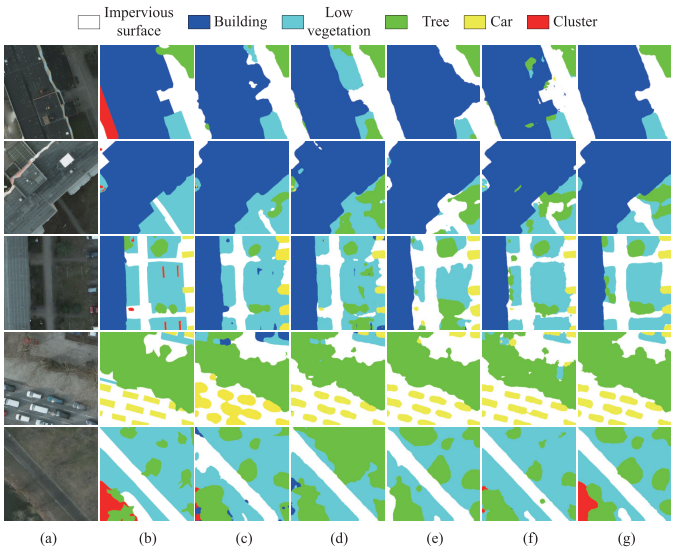


Fig. 6. Segmentation maps of different methods in the VH→PD task. (a) RSIs. (b) Ground truth. (c) ProDA. (d) CPSL. (e) Li's. (f) DAFormer. (g) PCEL.

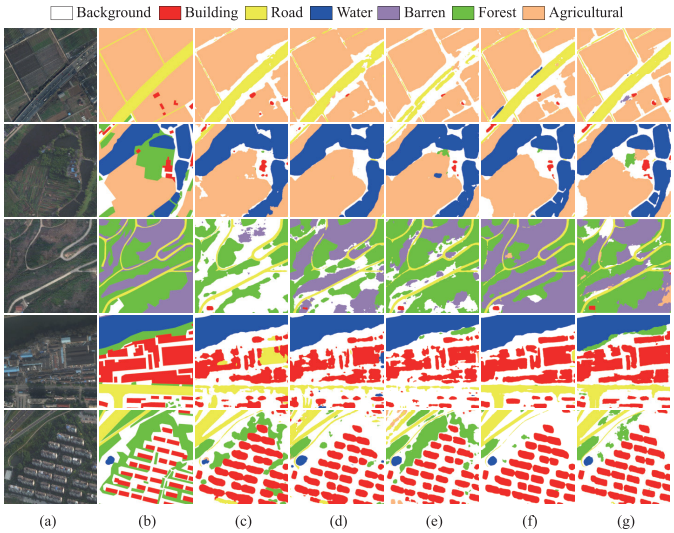


Fig. 7. Segmentation maps of different methods in the Rural→Urban task. (a) RSIs. (b) Ground truth. (c) ProDA. (d) CPSL. (e) Li's. (f) DAFormer. (g) PCEL.

line of Fig. 6(d). In contrast, the segmentation maps of Li's and DAFormer have fewer misclassified pixels and more complete spatial structure. However, the integrity of the segmentation results is still insufficient, and the ability to distinguish similar objects needs to be further improved. The proposed method can obtain the most best segmentation maps, and the improvement is mainly reflected in the following three aspects. First, the class confusion phenomenon is greatly reduced and the segmentation results between classes or instances are more accurate, which benefits from the cross-domain multiprototypes learning. For example, in the fourth line of Fig. 5 and the last line of Fig. 6, the proposed method can recognize the clutter class more accurately. Second, benefiting from the global consistency learning strategies, the segmentation results of objects with larger coverage have more complete structure and smooth contour. Third, as shown clearly in the red boxes of Fig. 5, the details in small regions are more accurately predicted, which benefits from the local consistency learning strategies.

E. Results of LoveDA Dataset

The LoveDA dataset is acquired by spaceborne sensors and has the wider ground coverage, which results in the larger domain discrepancies between the subsets Rural and Urban. Table IV lists the segmentation results of different methods in the Rural→Urban task, from which the obtained observations are generally consistent with those of Tables II and III. The adversarial learning methods Advent and Zheng's, as well as the RSI translation method RDG, have relatively poor performance and fail to produce satisfactory segmentation results. In contrast, the self-supervised learning methods, SAC, ProDA, and CPSL, can effectively improve the segmentation performance by optimizing the self-supervised learning process with different techniques and strategies. The two advanced transformer-based methods can make better use of long-distance dependencies and enhance the learning of rare classes, further improving the adaptation and transfer to target RSIs. In addition, it can be seen that the improvement effect brought by ProCST is not obvious in this domain adaptation task. The proposed method, PCEL, achieves the best segmentation performance according to the mIoU value, which increases by 0.94% compared to the second place. From the perspective of the segmentation results per class, the proposed method can achieve the comparable or even higher accuracy. In addition, the proposed method can effectively improve the accuracy of classes that are difficult to be recognized, such as agricultural, which benefits from the proposed cross-domain multiprototypes learning and masked consistency strategies.

Fig. 7 shows the segmentation maps to visually compare the results of different methods. It can be seen that the proposed method can produce the most accurate segmentation maps with better visual effect. Compared with other methods, the proposed method greatly reduces the error noises and class confusion in the segmentation maps. In addition, the proposed method has the better segmentation performance on the overall contextual structure of observation regions. Representative examples include the segmentation between narrow roads and square farmlands in the first line of Fig. 7, and the recognition of buildings in the last two lines of Fig. 7. In general, for the spaceborne RSIs with larger domain discrepancies, the quantitative and qualitative segmentation results of the proposed method are better than other methods.

V. Analysis and Discussion

A. Ablation Studies

The proposed method utilizes the learned cross-domain multiprototypes to model complicated class-level relations in RSIs domain adaptation and applies the masked consistency strategies in self-supervised training to further improve the adaptation and transfer to target RSIs. In this section, ablation studies are conducted to verify their respective effectiveness. The experimental settings and hyperparameters are consistent with Section IV-B. The results are presented in Table V, where CDMP is short for cross-domain multiprototypes and GCL and LCL are short for global consistency learning and local consistency learning, respectively.

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TABLE IV
SEGMENTATION PERFORMANCE OF DIFFERENT METHODS IN THE RURAL→URBAN TASK. IOU PER CLASS IS LISTED FROM COLUMN 3 TO COLUMN 9. BOLD VALUES INDICATE THE BEST RESULTS

Types	Methods	Venue	Background	Building	Road	Water	Barren	Forest	Agricultural	mIoU	PA
Adversarial learning	Advent	CVPR 2019	29.12	42.14	36.42	43.85	27.30	26.48	12.56	31.12	50.66
	Zheng's	TGRS 2022	43.73	37.22	32.22	48.92	21.26	**26.65**	10.97	31.57	52.69
RSI translation	RDG	RS 2023	49.58	36.17	36.58	55.73	19.07	15.68	15.37	32.60	53.85
Self-supervised learning	SAC	CVPR 2021	53.91	41.80	32.99	71.11	24.93	12.18	23.44	37.19	62.74
	ProDA	CVPR 2021	54.81	37.83	44.85	68.00	19.94	20.06	30.59	39.44	63.26
	CPSL	CVPR 2022	54.62	47.14	44.25	72.24	23.78	13.72	29.57	40.76	64.43
	Li's	RS 2022	53.75	37.24	45.67	68.49	36.76	12.10	30.99	40.71	63.30
	DAFormer	CVPR 2022	55.84	50.95	**49.95**	75.44	34.02	25.66	31.76	46.23	**67.10**
	DA-ProCST	arXiv 2022	**56.03**	50.87	49.16	74.74	35.81	25.97	32.52	46.44	67.02
	PCEL		54.19	**51.54**	47.83	**77.99**	**37.99**	23.80	**38.35**	**47.38**	66.57

TABLE V
ABLATION STUDIES

Tasks	Baseline	Baseline + CDMP	Baseline + CDMP + GCL	Baseline + CDMP + GCL + LCL
PD→VH	64.13	70.68	72.53	73.44
VH→PD	58.36	62.18	62.79	63.37
Rural→Urban	44.01	46.30	46.95	47.38

1) Effectiveness of Cross-Domain Multiprototypes: The baseline is actually the widely used self-supervised UDA method. After removing the cross-domain multiprototypes and masked consistency strategies, it can only utilize the cross-entropy function for loss calculation and can only utilize the information provided by pseudo labels for RSIs domain adaptation. It can be seen that the segmentation results of the baseline method are relatively poor, indicating that self-supervised learning alone could hardly achieve ideal performance in RSIs domain adaptation. After the addition of cross-domain multiprototypes, the segmentation results in the three UDA tasks have been significantly improved. Specifically, the mIoU values increase by 6.55%, 3.82%, and 2.29%, which directly indicates that the proposed cross-domain multiprototypes learning plays an important role in improving the performance of RSIs domain adaptation.

2) Effectiveness of Global Consistency Learning: Although the cross-domain multiprototypes can effectively improve the performance of RSIs domain adaptation, the exploration of baseline+CDMP on context structure of target RSIs is still limited. In fact, in addition to the improvement of domain alignment brought by modeling class-level relations, the baseline+CDMP method can only adapt to target RSIs using the information provided by pseudo labels. Therefore, the global masked consistency learning strategy is designed and applied, to further enhance the adaptation to target RSIs and the quality of pseudo labels through the learning of global context information. As we can see, the effectiveness of the global consistency strategy is particularly noteworthy in the PD→VH task, where the mIoU value increases by 1.85%. Also, in the other two tasks, the global consistency strategy also results in varying degrees of improvement. This indicates that exploring the global context information of target RSIs through the global consistency learning strategy can effectively improve the domain adaptation performance.

3) Effectiveness of Local Consistency Learning: Simply applying the global consistency strategy obviously neglects the learning of local information in target RSIs to a certain extent. In contrast, the adoption of local consistency strategy can further improve the performance of RSIs domain adaptation through optimizing the details of small targets and regions. Specifically, compared with the method that only adopts the global strategy, applying the two consistency learning strategies simultaneously can increase the mIoU values of the three tasks by 0.91%, 0.58%, and 0.43%. The above observations directly verify the effectiveness of different masked consistency learning strategies in the proposed method.

In summary, cross-domain multiprototypes learning is the key to improve the performance of RSIs domain adaptation in the proposed method, and the masked consistency strategies can further optimize the segmentation results on target RSIs.

B. Visual Analysis

In this section, two visual analyses are conducted to verify the effectiveness of cross-domain multiprototypes in modeling class-level relations. First, the distribution of the learned prototypes after the t-SNE [66] dimension reduction is shown in Fig. 8, from which three observations can be obtained.

1) In general, in the three different UDA tasks, the prototypes of different classes are separated from each other, while the prototypes within the same class are relatively close to each other, which can effectively model the interclass relations and improve the discrimination between different classes.
2) Within the distribution of a class, there is still a certain distance between different prototypes. Although the distance between some prototypes is relatively large, they are still located in the cluster of corresponding class. It can be considered that learning cross-domain multiprototypes can map intraclass change into the relative distance between different prototypes in the same cluster, so as to better describe rich intraclass relations.
3) Compared with the PD→VH and VH→PD tasks, the prototype cluster of each class in the Rural→Urban task is more dispersed. This can be attributed to the greater interdomain discrepancies and intradomain variations of spaceborne RSIs, so the distance between different prototypes within the same class is relatively larger.

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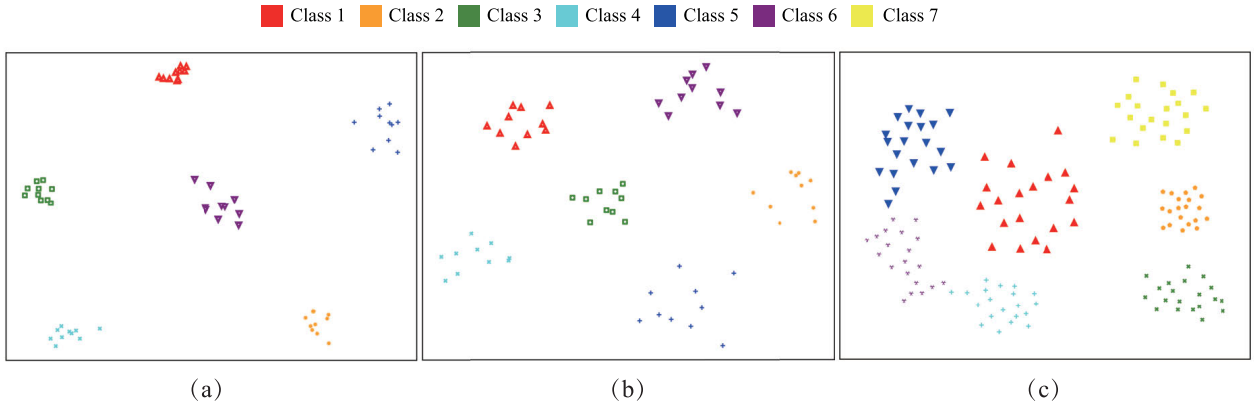


Fig. 8. Visualization of cross-domain multiprototypes with different colors representing different classes. (a) PD→VH. (b) VH→PD. (c) Rural→Urban.

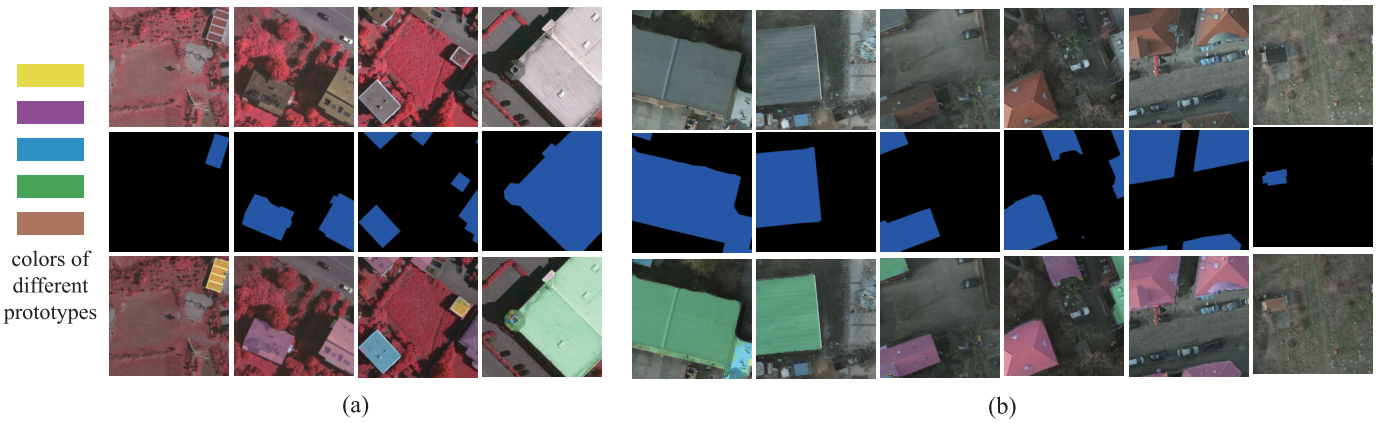


Fig. 9. Visualization of the corresponding relations between pixels and prototypes with the building class as an example in the VH→PD task. (a) VH. (b) PD.

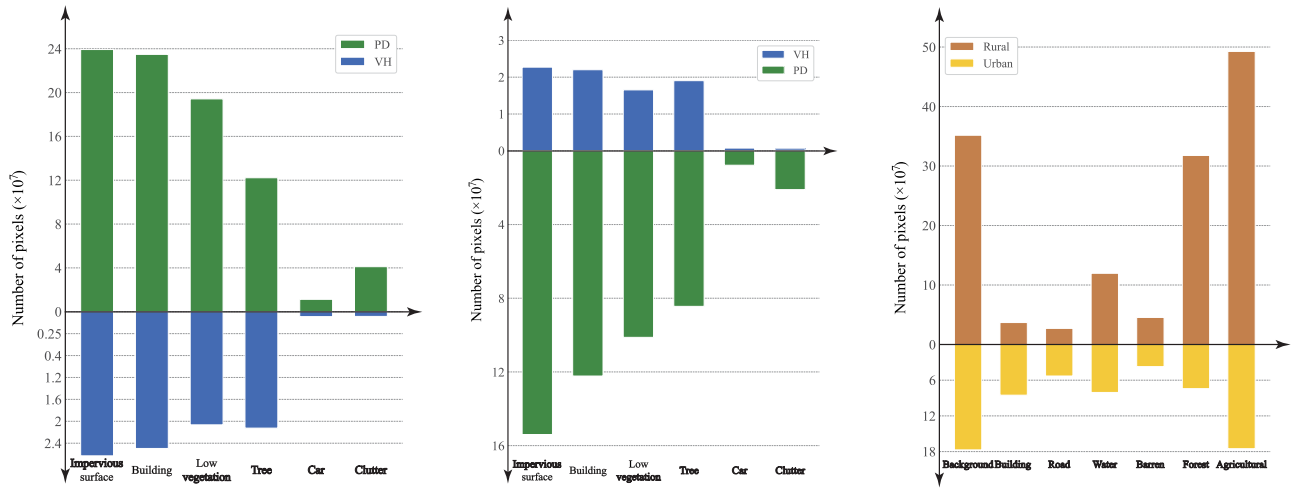
Next, the relations between pixels and prototypes are visualized and analyzed, with the building class as an example. Specifically, different colors are first assigned to different prototypes within the same class. Then, according to the assignment results, the prototypes obtained from label inference are displayed on the corresponding positions of pixels. In Fig. 9, the first row and the second row show the RSIs and their corresponding ground truths, respectively. The third row displays the weighted addition results of the colors of different prototypes and RSIs. As we can see, within the same class, the instances with great differences correspond to different prototypes, and the similar instances correspond to the same prototypes, whether across different RSIs domains or within the same RSIs domain. This directly verifies the representation ability of multiprototypes in establishing the intraclass relations. More importantly, the multiprototypes are learned on the common basis of source and target domains, that is, the same prototype has different instances corresponding to it in both domains, which can achieve better domain alignment and adaptation at the class level.

C. Class Imbalance Analysis

Considering the real distributions of ground objects in observation area, the class imbalance phenomenon is very common in RSIs. When semantic segmentation is carried out within the same RSI domain, this problem is mainly reflected in that different classes have a different number of pixels, which may lead to the inclination of the trained model to some majority classes. In domain adaptation learning, the problem of class imbalance is more complicated because of the imbalance between the same classes in different domains [67]. As shown in Fig. 10, the source and target domains themselves have the problem of class imbalance in the three domain adaptation tasks. Besides, the class distribution between the two domains is also quite different. Undoubtedly, the unbalance between domains and the unbalance between classes brings greater challenges to the cross-domain learning process, which will directly affect the segmentation performance on target RSIs. However, as far as we know, there are relatively few research works explicitly exploring the class imbalance problem in the domain adaptation segmentation of RSIs [8].

According to the statistical results in Fig. 10, the long-tailed classes of different datasets can be roughly identified, which include the last four classes in the ISPRS dataset and classes 2–5 in the LoveDA dataset. In fact, it can be seen from the IoU results in Tables II and IV that the proposed method performs better than other methods in these long-tailed classes, which is particularly evident in the classes of car, clutter, and tree in the ISPRS dataset, and the classes of barren and water in the LoveDA dataset. To further observe and analyze the ability of the proposed method to deal with the class imbalance problem in the process of domain adaptation, the confusion matrices of three different tasks are visualized in Fig. 11. The red box highlights the recognition performance of long-tailed classes, and the yellow box highlights the degree of

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(a) (b) (c)

Fig. 10. Illustrations of the unbalanced distribution of classes in three different tasks. (a) PD→VH. (b) VH→PD. (c) Rural→Urban.

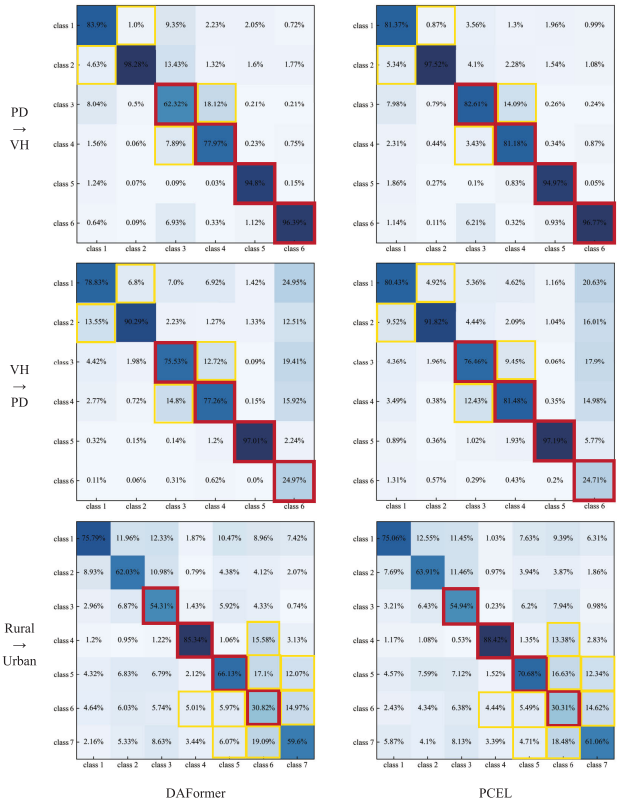


Fig. 11. Visualization of confusion matrices of three different tasks. The red box highlights the recognition performance of long-tailed classes, and the yellow box highlights the degree of misclassification between similar classes.

misclassification between similar classes. Different from the class sampling strategy adopted by DAFormer, the proposed method focuses on establishing the complicated class-level relations based on source and target samples, thus improving the segmentation accuracy of long-tailed classes. It can be seen that the proposed method is superior to DAFormer in the performance of long-tailed class recognition and similar class discrimination. This directly shows the effectiveness of the proposed method in dealing with the class imbalance problem in the domain adaptation segmentation of RSIs and verifies the advantages of the proposed cross-domain multiprototype learning strategy.

TABLE VI
RELATIONSHIPS BETWEEN THE NUMBER OF PROTOTYPES PER CLASS AND MIOU

Tasks	$K = 5$	$K = 10$	$K = 20$	$K = 40$
PD→VH	72.89	73.44	72.68	72.86
VH→PD	62.03	63.37	62.39	61.92
Rural→Urban	46.49	46.63	47.38	46.66

D. Hyperparameters Discussion

In this section, the influence of hyperparameters on segmentation results is analyzed and discussed in detail. First, the relationships between the number of prototypes per class and mIoU are presented in Table VI. As we can see, with the increase in the number of prototypes, the mIoU of three different tasks all shows a trend of first gradually increasing and then slowly decreasing. Obviously, a small number of prototypes cannot effectively capture rich class information, resulting in the decline of the performance of domain adaptation. Meanwhile, too many prototypes also weaken the representation effect while causing unnecessary computing. For the PD→VH task and VH→PD task, the optimal value of K is 10, while for the rural→urban task, the optimal value rises to 20. This could be attributed to the fact that the spaceborne RSIs with greater interdomain discrepancies and intradomain variances require more prototypes for modeling complex class-level relations.

Next, the relationships between the dimension of prototype and mIoU are explored, and the results are given in Fig. 12 The dimension of prototype could be understood as the complexity level required to abstractly model the instances of each subclass. As can be seen from the three curves, both too large and too small dimensions affect the segmentation performance of the proposed method on target RSIs to varying degrees. Specifically, the optimal value of dimension in the two tasks of the ISPRS dataset is 128, while the optimal value corresponding to the LoveDA dataset is 256. Similarly,

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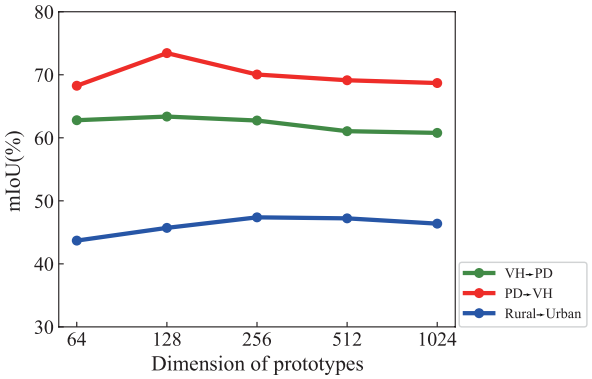


Fig. 12. Relationships between the dimension of prototype and mIoU.

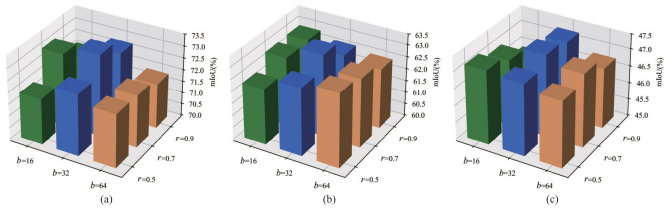


Fig. 13. Influence of the size of masked patches and mask ratio on mIoU. (a) PD→VH. (b) VH→PD. (c) Rural→Urban.

TABLE VII
INFLUENCE OF THE HYPERPARAMETER γ ON MIOU

Tasks	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$	$\gamma = 1.0$
PD→VH	71.05	71.73	72.41	72.55	73.44
VH→PD	62.43	62.65	63.09	63.02	63.37
Rural→Urban	46.53	46.46	46.87	47.02	47.38

it can be considered that, for the spaceborne RSIs with greater interdomain discrepancies and intradomain variances, a relatively larger dimension value can better represent the differences among instances.

Then, the influence of the spatial size of masked patches b and mask ratio r on mIoU is explored and analyzed, and the results are shown in Fig. 13. It can be seen that the proposed method has different performances under different combinations of b and r. On the whole, when r is determined, setting b to 32 can effectively improve the mIoU value. When b is determined, setting r to 0.7 tends to produce better results. Intuitively, in the process of masked consistency learning, setting an appropriate patch size and mask ratio can effectively promote the exploration of contextual structure of target RSIs, thus improving the effect of domain adaptation.

Finally, the optimal value of the hyperparameter γ is discussed, which determines the proportion of masked consistency learning in RSIs domain adaptation. Table VII shows the segmentation results of the proposed method when γ changes. When γ is too small, the masked consistency strategy makes little contribution to domain adaptation learning, and the proposed method has poor segmentation performance. As the value of γ increases, the mIoU value in Table VII increases gradually. The best segmentation results can be obtained when γ is 1. This directly verifies the effectiveness of the masked consistency strategy in improving RSIs domain adaptation.

VI. CONCLUSION

In this article, a novel PCEL method is proposed based on the framework of self-supervised domain adaptation, which can further improve the performance of UDA semantic segmentation of RSIs from both the two aspects of prototype and context. Learning cross-domain multiprototypes for each class can abstract the complex interclass and intraclass relations into multiple prototypes, to effectively alleviate the difficulties brought about by huge interdomain discrepancies and intradomain variances for RSIs domain adaptation. In addition, the cooperative participation of source and target samples can make the learned prototypes contain rich feature information of different RSIs domains. Considering the objects of different scales in RSIs, the masked consistency learning strategies can effectively enhance the learning of global context relations and multiscale local context details, thus optimizing the domain adaptation process and improving the quality of pseudo labels used for prototype updating. Through the enhancements of prototype and context, the proposed method can effectively improve the adaptation and transfer to target RSIs. Experimental results on airborne and spaceborne RSIs demonstrate that, compared with existing advanced UDA methods, the proposed method can achieve better segmentation performance on target RSIs.

Although the performance of UDA semantic segmentation of RSIs is effectively improved, the proposed method still requires separate domain adaption training for each target RSIs. In the following work, the domain generalization methods will be introduced to further improve the adaptability of the model to different RSIs domains.

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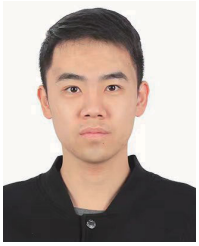
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